

This is due by July 4.

Problem 1. (10 points.) Let $n \geq k \geq 0$ be integers and define $\binom{n}{k} = \frac{n!}{(k!)(n-k)!}$.

1. Show $\binom{n}{k} + \binom{n}{k+1} = \binom{n+1}{k+1}$. For a given integer $d > 0$, compute $\sum_{i=0}^n \binom{i+d-1}{i}$.
2. For integers x, y , show $(x+y)^n = \sum_{i=0}^n \binom{n}{i} x^i y^{n-i}$.

Problem 2. Find the smallest positive integer in the following sets where $a = 60$ and $b = 54$. Explain why the integer you find is the desired one.

1. $\{na - mb : n, m \in \mathbb{Z}\}$
2. $\{a - mb : m \in \mathbb{Z}\}$
3. $\{3na - mb : n, m \in \mathbb{Z}\}$

Proof. 1. This is the greatest common divisor of a and $-b$, which equals $(a, b) = 6$.

2. First we note that $6 = 60 - 54$ which lies in $\{a - mb : m \in \mathbb{Z}\}$. In addition, the set

$$A = \{a - mb : m \in \mathbb{Z}\} \subseteq \{na - mb : n, m \in \mathbb{Z}\} = B.$$

In particular, the smallest positive integer in A is no smaller than the smallest positive integer in B . But we have seen that the smallest positive integer in A is that in B . So it must be 6.

3. This is $(3a, -b) = (3a, b) = 9$. □

Problem 3. Let $a, b, c \geq 0$ be integers and consider the set $(a, b, c) = \{ax + by + cz : x, y, z \in \mathbb{Z}\}$.

1. Show there exists some $d \in (a, b, c)$ such that $d \mid r$ for all $r \in (a, b, c)$.
2. Find three particular choice of a, b, c such that the set $T = \{a - nb - mc : n, m \in \mathbb{Z}\}$ does not contain an element d such that $d \mid r$ for all $r \in T$.

Proof. 1. Consider $R = (a, b, c) \cap \mathbb{N}_+$. Note $R = \emptyset$ iff $a = b = c = 0$ and $(a, b, c) = 0$. The let $d = 0$. Now assume $R \neq \emptyset$. Let $d = \min R$. We claim $d \mid r$ for all $r \in (a, b, c)$. By the division algorithm, write $a = q_1 d + r_1$ for $0 \leq r_1 < d$. Note that $r_1 = a - q_1 d \in R$. So $r_1 = 0$ otherwise it contradicts the minimality of d . This implies $d \mid a$. Similarly $d \mid b$ and $d \mid c$.

Fred: Directly apply the division algorithm and write $r = dq + r'$ for $0 \leq r' < d$ for arbitrary $r \in (a, b, c)$ and conclude $r' = 0$.

2. Let $a = 2, b = 5, c = 0$. Then $2 \in T$ and $-3 \in T$. But $(2, -3) = 1$ which is not in T . □

Problem 4. Let a, b, c and n be positive integers. Show the following.

1. If $n \mid a$ and $n \mid b$, then $n \mid ax + by$ for all $x, y \in \mathbb{Z}$.
2. If $ab \mid c$, then $b \mid \frac{c}{(a,c)}$.
3. $n(a, b) = (na, nb)$. (Give an counterexample when n is not positive.)
4. $(\frac{a}{(a,b)}, \frac{b}{(a,b)}) = 1$
5. If $ca \equiv cb \pmod{n}$, then $a \equiv b \pmod{\frac{n}{(n,c)}}$.

proof of part 3. Let $(a, b) = d$ and $(na, nb) = d'$. Then $nd \mid na$ and $nd \mid nb$. So $nd \mid d'$ which implies $nd \leq d'$. Conversely, note n is a common divisor of na, nb . So $n \mid d'$. That is d'/n is really an integer. But then $d'/n \mid a$ and $d'/n \mid b$ since we can divide n from $d' \mid na$ and $d' \mid nb$. So $d'/n \mid d$ which implies $d'/n \leq d$.

When $n = -1$, the statement is not true for the LHS is negative and the RHS is always positive.

Albert: Note that (na, nb) is the least positive integer in the set $\{xna + ynb : x, y \in \mathbb{Z}\}$. Since n is positive, a positive element $xna + ynb = n(xa + by)$ is the least only when $xa + by$ is the least, which is (a, b) . $\square$

proof of part 4. Suppose $(\frac{a}{(a,b)}, \frac{b}{(a,b)}) = d > 1$. Then $d(a, b) > d$ is a common divisor of a and b , contradicting d is the greatest common divisor. $\square$

Problem 5. Let $a, b, d > 0$ be integers.

1. Show that there exist $x, y \in \mathbb{Z}$ such that $ax + by = d$ if and only if $(a, b) \mid d$.
2. Let $a = 13$ and $b = 37$. Find one pair of integer solution (x_0, y_0) to the equation $ax + by = 1$.
3. Suppose $(a, b) \mid d$ and (x_0, y_0) is a pair of integer solution to the equation $ax + by = d$. Show that $(x_0 + k \cdot \frac{b}{(a,b)}, y_0 - k \cdot \frac{a}{(a,b)})$ where $k \in \mathbb{Z}$ are all the integer solutions.

proof of part 3. First, it is easy to verify those are solutions to the above equation. Second, suppose (x, y) is another solution. Then $ax + by = d = ax_0 + by_0$ which implies $a(x - x_0) = b(y_0 - y)$. Dividing (a, b) from both sides, we get $\frac{a}{(a,b)}(x - x_0) = \frac{b}{(a,b)}(y_0 - y)$. Now as $(\frac{a}{(a,b)}, \frac{b}{(a,b)}) = 1$ and the Euclid's Lemma, we conclude that $\frac{a}{(a,b)} \mid y_0 - y$ and $\frac{b}{(a,b)} \mid x - x_0$. So we write $x - x_0 = k \frac{b}{(a,b)}$ and $y_0 - y = l \frac{a}{(a,b)}$ for some $k, l \in \mathbb{Z}$. Plug them back in the equation, it is easy to see $k = l$. So $x = x_0 + k \frac{b}{(a,b)}$ and $y = y_0 - k \frac{a}{(a,b)}$ are all the solutions. $\square$